RR-W-456D
December 18, 1968
SUPERSEDING
Fed. Spec. RR-W-456C
April 14, 1965

FEDERAL SPECIFICATION

WIREWARE, KITCHEN AND BAKESHOP:

POT CLEANERS, POTATO MASHER, SKIMMER, AND EGGWHIPS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers both domestic and institutional kitchen and bakeshop wireware.

1.2 Classification. The kitchen and bakeshop wireware shall consist of the following items and sizes, as specified (see 6.2).

Item 2 - Pot cleaner, metal link, without handle.
Item 3 - Pot cleaner, metal link, with handle.
Item 5 - Masher, potato, hand.
Item 6 - Skimmer, Kitchen.
Item 7 - Eggwhip.

Size 1 - 10 inches long by 2 inches diameter (min.).
Size 2 - 12 inches long by 2-1/2 inches diameter (min.).
Size 3 - 16 inches long by 3 inches diameter (min.).
Size 4 - 20 inches long by 3-1/2 inches diameter (min.).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 7330, 7920

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Federal Specifications:

L-P-394	-	Plastic Molding Material, (Propylene Plastics, Injection and Extrusion.)
QQ-N-281	-	Nickel-Copper Alloy - Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, And Structural And Special Shaped Sections.
QQ-W-423	-	Wire, Steel, Corrosion-Resisting.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), And Cut Shapes.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized And Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Sanitation Foundation (NSF) Publication:

Basic Criteria C-2 - The Evaluation of Special Equipment and/or Devices.

Approval Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitation Foundation, P. O. Box 1468, Ann Arbor, Michigan 48106.)

American Society for Testing and Materials (ASTM) Standard:

A511 - Seamless Stainless Steel Mechanical Tubing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19106.

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), before production is commenced, a sample item ordered under this specification shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The items delivered under this specification shall be the manufacturer's commercial standard product except for any changes necessary to comply with specification requirements.

3.3 Standards. The potato masher, the skimmer, and the size 1 and 2 eggwhips furnished under this specification shall conform to the applicable requirements of NSF Basic Criteria C-2.

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3.3.1 NSF certification. The supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the potato mashers, the skimmers, and the size 1 and 2 eggwhips he proposes to furnish under this specification meet the applicable requirements of NSF Basic Criteria C-2 as follows:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished items shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to the approval of the first shipment, or:

(2) A certified test report from a recognized independent testing laboratory acceptable to the medical authority functioning in advisory capacity to the preparing acitivity of the specification indicating that the items have been tested and conform to NSF Basic Criteria C-2 shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.4 Materials. Materials not definitely specified shall be of the quality normally used for the purpose in the manufacturer's current product. Unless otherwise specified herein, commercial tolerances are applicable to material thicknesses and diameters.

3.4.1 Wire.

3.4.1.1 Corrosion-resisting steel. Corrosion-resisting steel wire shall conform to the chemical and mechanical requirements of composition 302 or 304, of QQ-W-423, except for tensile strength specified for potato masher and eggwhips.

3.4.1.2 Carbon steel. Carbon steel wire shall be bright, hard drawn, commercial quality.

3.4.1.3 Nickel-copper alloy. Nickel-copper alloy wire shall conform to class A of QQ-N-281.

3.4.2 Molding plastic. Molding plastic material shall be polypropylene conforming to type I or III, grade A or B of L-P-394.

3.4.3 Corrosion-resisting steel tube. Corrosion-resisting steel tube shall be round and shall conform to grade MT-302 or 304 of ASTM A511.

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3.5 Design and construction. The wireware items shall conform to the applicable requirements specified herein and shall be similar to figures 1, 2, 3, or 4 as applicable. Figures 1, 2, 3, and 4 are for illustrative purposes only and are not intended to preclude procurement of items otherwise meeting all the applicable requirements of this specification.

3.5.1 Item 2, pot cleaner, metal link, without handle. Item 2 pot cleaner (see figure 1) shall be fabricated of the wire specified in 3.4.1.2 with a diameter of not less than 0.054 inch. The pot cleaner shall consist of a double mesh of interlocking rings and the overall dimensions shall be not less than 5 by 7-1/2 inches when laid flat. The wire rings shall be not less than 1/2 inch diameter and each ring shall have not less than 1-1/2 turns of wire.

3.5.2 Item 3, pot cleaner, metal link, with handle. The mesh of the Item 3 pot cleaner shall be the same as specified in 3.5.1, except that when laid flat it shall be not less than 6-1/2 inches diameter. The handle shall be not less than 4-1/2 inches in length and the base of the handle shall form a scraper of not less than 2 inches in width.

3.5.3 Item 5, masher, potato, hand. The potato masher (see figure 2) shall be not less than 12 inches overall in length and the base of the masher shall be not less than 3-1/8 by 4-1/4 inches. The wire forming the base and the lower part of the handle shall be as specified in 3.4.1.1 with a diameter of not less than 0.192 inch and a tensile strength of not less than 120,000 pounds per square inch (p.s.i.). The wire shall be bent in a serpentine shape and the ends of the wire bent at right angles to the base, brought together and inserted into a handle. The handle shall be tubular and fabricated of corrosion-resisting steel, brass, carbon steel, or shall be molded of the plastic specified in 3.4.2.

3.5.4 Item 6, skimmer, kitchen. The skimmer (see figure 3) shall be not less than 1/2 inch deep by 4 inches in diameter and shall have a handle not less than 9 inches in length. The skimmer shall be fabricated of the wire specified in 3.4.1.1. The frame wire shall be not less than 0.1205 inch diameter. The filler wires shall be not less than 0.041 inch diameter and shall be attached to the frame by at least one cross wire. The filler wire shall consist of not less than 11 evenly spaced concentric rings or of a single spiral of not less than 11 complete loops. The frame wire shall form the skimmer circumference and may be bent to form the handle, or bent and inserted into a metal or a molded plastic handle constructed of material specified in 3.4.2. The handle may also be formed of the same diameter wire as the frame and welded to the frame.

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3.5.5 Item 7, eggwhips. The eggwhips (see figure 4) shall be size 1, 2, 3, or 4 and shall be of the dimensions specified in 1.2 and as specified herein. Each eggwhip shall have not less than 8 bow wires fastened to a tubular metal handle or a molded plastic handle. The tubular metal handle shall be sealed at the end opposite the bow wires and the bow wires shall be held in place and sealed in the metal handle with a poured solder plug, or a corrosion-resisting steel or tinned metal insert. The bow wires shall be held in place and sealed in the molded plastic handles in accordance with the manufacturers standard practice.

3.5.5.1 Sizes 1 and 2. The bow wires for size 1 and 2 shall be fabricated of wire specified in 3.4.1.1 with a tensile strength not less than 200,000 p.s.i.; wire specified in 3.4.1.2; or wire specified in 3.4.1.3 and shall have a diameter not less than 0.080 inch. Handles shall be fabricated of carbon steel tube, brass tube, or corrosion-resisting steel tube specified in 3.4.3, or molded of the plastic specified in 3.4.2. Metal handles shall have an inside diameter of not less than 0.875 inch, a wall thickness of not less than 0.020 inch, and shall be 4-1/2 inches plus or minus 1/2 inch in length. Molded plastic handles shall be 4-3/4 inches plus or minus 1/4 inch in length and not less than 1-1/8 inches in diameter at the bow wire end.

3.5.5.2 Size 3. The bow wires for the size 3 eggwhip shall be fabricated of corrosion-resisting steel wire specified in 3.4.1.1 with a tensile strength not less than 200,000 p.s.i. and with a diameter not less than 0.1055 inch. The handle shall be fabricated of corrosion-resisting steel tube specified in 3.4.3 or molded of the plastic specified in 3.4.2. Handle dimensions shall be as specified in 3.5.5.1. The space between bow wires at the handle shall be not less than 1/16 inch.

3.5.5.3 Size 4. The bow wires for the size 4 eggwhip shall be fabricated of wire specified in 3.4.1.1 with a tensile strength of not less than 200,000 p.s.i.; wire specified in 3.4.1.2; or wire specified in 3.4.1.3 and shall have a diameter not less than 0.1055 inch. The handle shall be fabricated of carbon steel tube, brass tube, or corrosion-resisting steel tube specified in 3.4.3 or molded of the plastic specified in 3.4.2. Metal handle dimensions shall be as specified in 3.5.5.1. Molded plastic handles shall be 5-1/2 inches plus or minus 1/4 inch in length and not less than 1-1/4 inches in diameter at the bow wire end. The spacing between bow wires at the handle end shall be not less than 1/16 inch.

3.6 Marking for identification. Marking for identification shall be in accordance with the manufacturers standard commercial practice.

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3.7 Finish. All components fabricated of the carbon steel wire specified in 3.4.1.2, except the pot cleaners, shall be tinned and retinned or nickel and chromium plated. Handles of carbon steel or brass shall be nickel and chromium plated in accordance with the manufacturer's standard practice.

3.8 Workmanship. The finished items shall be clean and free of imbedded foreign matter, rust, sharp edges, burrs, slivers, die marks, deformities, fractures, scale, and pits. When poured solder plugs or tinned metal inserts are used in the eggwhips, excess solder or tin shall be removed from the bow wires and handles. There shall be no sharp exposed ends on the wire loops of the pot cleaners. Plated and tinned finishes shall cover all areas of the specified components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required, it shall be examined for the defects listed in table I and dimensions specified.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End item inspection. The lot shall consist of all like items, or all like items of the same size, as applicable, offered for inspection at one time. The sample unit shall be one completed fabricated item.

4.3.2.1 Visual inspection. Inspection shall be made of the items for the defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

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TABLE I. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not plated or tinned (where required)	X	
	Areas of no coverage, blisters, or peeling	X	
	Dirty		X
	Imbedded foreign material	X	
	Rust, scale, or pitts	X	
Design, construction and workmanship, (general)	Component missing or not as specified	X	
	Slivers, die marks, deformites, fractures, or sharp edges	X	
Pot cleaners (with or without handles)	Open rings, or sharp ring ends	X	
	Rings less than 1-1/2 turns of wire		X
Pot cleaner with handle	Handle not attached to mesh	X	
	Handle base not formed as a scraper	X	
Masher, potato	Base wire not shaped as specified	X	
Skimmer	Filler wires not fastened to frame by at least one cross wire	X	
	Less than 11 concentric circles or less than 11 complete loops of filler wire	X	
Eggwhips	Less than 8 bows of wire	X	
	Bow wires not held in place in metal handles with a poured solder plug, or a corrosion-resisting steel or tinned metal insert	X	
	Bow wires not sealed in metal handles	X	
	Excess tin not removed from bow-wires and handle (when applicable)		X
	Tubular metal handles not sealed at ends opposite bow wires	X	

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4.3.2.2 Dimensional examination. An examination shall be made to determine compliance with dimensions specified. Any noncompliance shall be considered a defect. The inspection level shall be S-3 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.3 Standards compliance. Proof of compliance with the requirements of 3.3 shall be made available to the Government representative.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in Table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examination	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Contents (interior)	More or less than required.
Weight	Net weight exceeds requirements.

4.3.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Wireware of one item and size only shall be packaged in the quantitites specified in table III in a snug-fitting fiberboard box conforming to type CF, variety optional, or type SF, class domestic of PPP-B-636. Each box shall be securely closed in accordance with the appendix of PPP-B-636.

5.1.2 Level C. The wireware shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

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TABLE III. Packaging and packing

Item	Quantity per box	Quantity per pack
Pot cleaner, metal link, no handle	36	(see note 1)
Pot cleaner, metal link, with handle	24	(see note 1)
Masher, potato, hand	12	60
Skimmer, kitchen	12	72
Eggwhip, size 1	12	72
Eggwhip, size 2	12	72
Eggwhip, size 3	12	72
Eggwhip, size 4	12	72

NOTE 1: The weight of contents for each pack shall not exceed 65 pounds.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Wireware of one item and size only, packaged as specified in 5.1, shall be packed in the quantities specified in table III in a snug-fitting fiberboard shipping container conforming to V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Wireware of one item and size only, packaged as specified in 5.1, shall be packed in the quantities specified in table III in a snugfitting fiberboard shipping container conforming to type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2) the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. Wireware, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

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5.3 Palletization. Unless otherwise specified (see 6.2) wireware packed as specified shall be palletized in accordance with Load Type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The wireware covered by this specification is intended for use in permanent kitchens and galleys and field and mobile kitchens.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Item number, or item number and size required (see 1.2).
(c) First article. When a preproduction sample is required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(f) When palletization is not required (see 5.3).

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 84

User activities:

Navy - MC, CG

Review activities:

Army - MD
Navy - SA, MS
Air Force - 84

Preparing activity:

Army - GL

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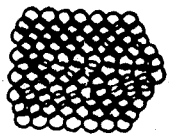


FIGURE 1.

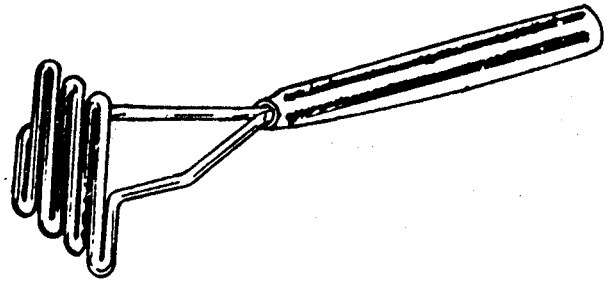


FIGURE 2.

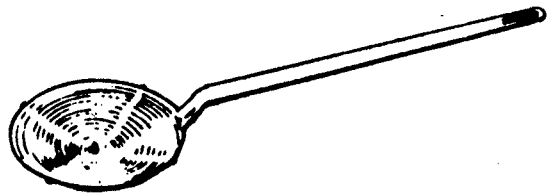


FIGURE 3.

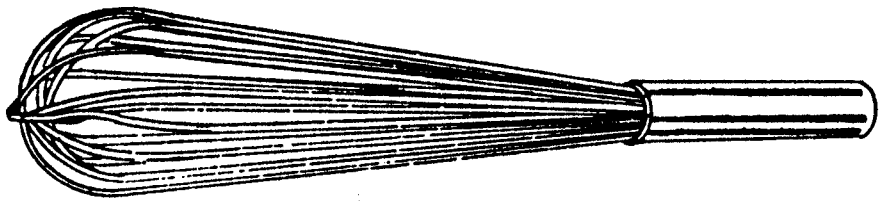


FIGURE 4.

Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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